

Schematic completion status

PetSafe 100-1339 R03 A | v0.9.9 | 2026-10-08

All catalog entries are drawn on one sheet. The remaining questions are listed explicitly below. This is a partially verified reconstruction, not a completed hardware validation or a routed PCB.

Coverage	Open connections	Values / packages
150 entries; 1 A2 sheet; 20 stock + 4 custom symbol definitions.	10 populated-entry pads + 18 DNP pads. 0 open PIC pads in the model.	44 ceramic values and L1/L2 unknown; 147/150 footprints assigned.

Power rails - current names

Name	Known path
BATTERY+	Battery positive -> Q1 single pad/native3 (candidate drain); TP14/R43.2 on the raw side.
VSYS	Q1.L/native2 -> U5/MX512H pin4; L1.1 and L2.1 supply sides.
VDD	TP103, PIC20/C32, U5 pin1/VCC, V064/V063/V075/V082.
VREF	TP104, PIC21/RB0/V026, PIC4/RA2/V035/C39, R21.2/V071.
GND	Battery negative and ground copper.

L1/L2 share the VSYS supply side. Their opposite ends are not shorted together. H_RF_VDD, H_LDO_IN, H_RX_VDD and H_PIR_VDD describe local branches; these are not additional confirmed independent supplies. Rail voltages have not been measured.

Recommended next checks

1. Identify which D3 pad joins V082/VDD (E02).
2. Resolve D2, C6/Q8, and the isolated button/antenna paths.
3. Cross-check receiver/PIR branches and candidate pin functions.
4. Recover RF/tuning values, then other passives and missing package geometry.

Validation and limits

83 modeled net partitions match KiCad 10.0.5. ERC retains 36 findings: 28 open pins, 3 isolated labels and 5 undriven power checks. No artificial NC/power flags hide gaps. All 26 original photo hashes are unchanged. This proves file consistency, not all physical connections.

The three isolated labels are H_ANT_B (ANT2), H_D1_FREE (D1.R) and H_BUTTON_MCU (R41.1). A PIC pin can have a local net and still lack an onward destination. Question-mark count is not a completion metric.

Priority circuit gaps

E01 - D2 / marking 4P

Known: The user clearly reads 4P; its three physical pads are retained by a numbered stock placeholder.

Uncertain: Identity, internal pin functions and all three external connections. The earlier BAV99 clamp assumption is withdrawn.

Next: With power disconnected, take the six directed diode-mode readings between D2 pads shown in the existing D2 guide. Then trace only the terminal indicated by the result.

E02 - D3 / V082 supply association

Known: A7 supports a BAV99 candidate. User continuity joins V082 to the board VDD test point.

Uncertain: Which physical D3 terminal reaches V082. Current D3.L=GND, D3.R=H_RX_VDD and D3.S=H_RX_RF_SENSE are provisional; none is presently assigned to VDD.

Next: Measure V082 to each of D3's three pads. Record low-ohm continuity separately from diode/resistive readings.

E03 - RF supply / Q8 / C6

Known: Q8 centre B2/native5 is GND via V113/V115. V114 is U2.T2/native2 GND. Outer Q8 pads are joined by the earlier low-resistance result.

Uncertain: C6.1 route and Q8 source-supply path. F-P=10 ohm supports a supply-related path but does not establish a direct copper join. Exact Q8 maker remains a candidate.

Next: Start with C6's non-ground pad and the L1 output/Q8 supply node; record resistance both ways. Use the existing labelled Q8 photo, not the withdrawn C6-to-output wire.

E04 - Antenna / five tuning cells

Known: All five PIC control inputs are mapped. Seven capacitor-midpoint comparisons gave OL.

Uncertain: Remote paths of the five capacitor midpoints and ANT2; the modeled tuning network is not yet a verified closed antenna circuit. Transistor orientation is provisional.

Next: Follow one cell at a time from ANT1/ANT2 toward a visible capacitor pad or via. Choose the next specific pair from the photographs; do not repeat a broad midpoint sweep.

E05 - Button circuit

Known: R41=331 gives nominal 330 ohm. One modeled button side is GND.

Uncertain: R41.1/H_BUTTON_MCU has no destination. S1 common lead pairs are currently assumed from a normal tactile switch.

Next: With power disconnected, determine S1 common pairs released/pressed; then trace R41.1 toward the MCU side.

E06 - Local-only PIC / discrete nodes

Known: Every PIC pad has a modeled local net. PIC2 reaches C25/V031; PIC14 reaches TP11; PIC18 reaches TP17.

Uncertain: Onward roles of those three nodes, D1.R/H_D1_FREE, C26.1 and TP9.1. Local connectivity is not proof that these are intentionally unused.

Next: Inspect the matching opposite-side vias first, then request only a specific continuity pair for each unresolved branch.

Other routing and identity gaps

E07 - PIR interface / U6 / J3

Known: C2NM supports an S-812C33AMC candidate. V067 reaches J3.1/red and the U6A option pad; V063 reaches regulated VDD. U6 ground is supported by the user readings.

Uncertain: U6 VIN/VOUT external branches, NC-pad external ties, actual PIR voltage, J3.2/.3 roles and Q2 signal path. The model uses J3.2=GND and J3.3=raw signal; blue/black colour alone is not confirmation or grounds to swap them.

Next: First verify J3.2 and J3.3 against GND and the modeled signal endpoint; then check U6 supply/output continuity. Powered voltage checks come after this routing audit.

E08 - Receiver / bias / Q7

Known: PIC12/RC1 reaches R18 via V053-V070. V064-V075 is continuity; V064-V070=11.2 kohm supports the R18+R19 path. VREF reaches R21.2, PIC21 and PIC4. TP6/C18.2 is separate from U3.5/C18.1.

Uncertain: Q7 terminal selection and power-switch role, receiver feedback/coupling branches, and R22.2/R23.2/R48.2 VREF ties remain inferred. TP6/V072 and U3.5/V079/V080 lack fully verified onward roles.

Next: Resolve E02 first because it may change receiver power. Then confirm the three provisional VREF bias branches and selected U3 feedback endpoints; retain C18 as a component boundary.

E09 - Power input / Q1 / rail voltages

Known: BATTERY+ and VSYS are separate. User confirms Q1.L to U5.4 and both L1/L2 supply sides on VSYS. VDD TP is the separate logic rail. R1A matches FMOS3401A marking.

Uncertain: Exact Q1 maker and gate operation, unmeasured rail voltages, and U1 enable/option branches. Reverse-polarity protection is an interpretation, not a powered test.

Next: Retain the confirmed rail split. Cross-check the Q1 gate/R44 path and U1 enable wiring before a targeted voltage test.

E10 - ICSP protection / D6 / J1

Known: D4/D5=5U supports SD05 TVS candidates. D6=G3 has at least two documented Zener candidates. ICSP signal routes are photo-derived.

Uncertain: Both D6 pad routes, D6 exact voltage/polarity, and J1 physical pin-number orientation including square-pad/VPP correspondence.

Next: Measure D6 pads to the adjacent PIC/ICSP nodes. Confirm J1 pad order against the board labels before relying on connector numbering.

I01 - Candidate identities / transistor pin maps

Known: Primary datasheets and marking evidence are recorded; custom symbols use the selected candidate pin tables.

Uncertain: Short markings do not uniquely prove manufacturers. Candidate choices and some B/E/C or MOSFET source/drain assignments remain qualified. Western alternatives are redesign options, not identities of fitted parts.

Next: Use electrical pad behavior and package/markings evidence for specific candidates. Do not request sharper photos where the existing marking is already legible.

I02 - LED / motor connector

Known: V127 reaches the photo-selected LED common pair; the two PIC LED drives are mapped. MX512H OUTA/OUTB map to the modeled motor connector.

Uncertain: LED internal polarity/colour mapping and J5 physical A/B lead polarity remain provisional.

Next: Check LED pad pairs in diode mode and record emitted colour. Match J5 wire positions to U5 outputs by continuity.

Values, mechanics and scope

V01 - Unmarked passive values

Known: C5 is 470 uF / 16 V; 38 fitted resistors have nominal marked values. R8 full-body dimensions support 0603 and C11 dimensions support 1206.

Uncertain: 44 fitted ceramic capacitances and L1/L2 type/value remain unrecovered. Capacitor voltage rating/dielectric and resistor tolerance/power rating are also not established by geometry.

Next: Use the LCR meter first on C6 and one tuning cell, recording frequency/mode and in-circuit status. Isolate one terminal only if an individual value is needed and the network reading cannot resolve it.

M01 - Package geometry / PCB reproduction

Known: 147 of 150 catalog entries have candidate stock footprints. R8 and C11 have user dimensions.

Uncertain: C5, S1 and LED1 footprints are blank. Other families, connector pitch, test-pad drills, switch lead span and exact board geometry are partly photo-based.

Next: Measure C5 can diameter/lead spacing, S1 body/lead span and LED1 body/pad pitch. Refine other footprints only where a photo scale or ruler contradicts the selected family.

D01 - Empty options

Known: 18 open pads belong to unpopulated options, not missing fitted chips.

Uncertain: Optional-pad routing is incomplete; values and intended uses may remain unknowable without another board variant.

Next: Record option copper where visible; defer measurement unless it affects a populated net.

F01 - Function / internal layers / daughterboard

Known: A one-sheet schematic and photo/via evidence exist; 83 modeled net partitions match the native schematic.

Uncertain: No powered functional validation, actual layer stack or routed KiCad PCB has been created. The PIR daughterboard internal circuit is not reconstructed. MCU firmware behavior and operating timing are not established.

Next: Finish priority electrical gaps first. Then test defined functions/voltages and decide whether physical PCB reproduction and daughterboard reverse engineering are needed.

Every unrecovered ceramic value

C1, C2, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48

C5 is already 470 uF / 16 V. In-circuit LCR readings may include parallel components and semiconductor paths; record frequency, mode and whether a lead was isolated. Geometry cannot establish capacitance, dielectric or voltage rating.

PIC connections

Physical package pins. All have modeled local nets; the evidence column distinguishes confirmed endpoints from photo-derived or incomplete branches. Use the separate numbered photo guide for physical locations.

Pin	Function	Local destination / evidence
1	RE3/MCLR	J1 VPP; mapping inferred; D6 route missing
2	RA0	C25 signal pad / V031; onward destination unknown
3	RA1	R28 + R29 + TP7; user meter / visible trace
4	RA2	VREF TP via V035; user confirms via-to-TP; C39.1 local photo route
5	RA3	V036-V108 to R14; user continuity report
6	RA4	V037-V102 to R15; user continuity report
7	RA5	V039-V097 to R16; user continuity report
8	VSS	GND / V041 photo anchor
9	RA7/OSC1	Y1 oscillator; photo-derived
10	RA6/OSC2	Y1 oscillator; photo-derived
11	RC0	V047-V091 to R11; user continuity report
12	RC1	R18.1 via V053-V070; user-confirmed; VREF link explicitly rejected
13	RC2	TP3 via V049; user-confirmed; existing R10 branch photo-derived
14	RC3	TP11; visible route; onward destination unknown
15	RC4	V022-V016 to U5 pin2 / INA; user-confirmed
16	RC5	V023-V019 to U5 pin3 / INB; user-confirmed
17	RC6	R13; user meter / visible trace
18	RC7	TP17; visible route; old R13 guess withdrawn
19	VSS	GND / C32 / V045 photo anchor
20	VDD	V042 to board VDD TP user-confirmed; C32.1 local photo route
21	RB0	VREF (TP104), V026 and V071/R21.2; user-confirmed
22	RB1	TP2 / R31; user meter / visible trace
23	RB2	TP1 / R30; user meter / visible trace
24	RB3	R4; photo-derived; sensing role remains inferred
25	RB4	TP4 / V017 / R42.1 / R44.1; user continuity; C41 photo-derived
26	RB5	TP10 / R35; user meter / visible trace
27	RB6/CLK	ICSP CLK / D4 / R32; photo-derived
28	RB7/DAT	ICSP DAT / D5 / R40; photo-derived

Every remaining open pad

Populated entries: C26.1, C6.1, D2.L, D2.R, D2.S, D6.1, D6.2, TP9.1, U6.L1, U6.L2.

Empty options: C42.1, C42.2, C49.1, C49.2, J6.1, J6.2, Q9.S, R36.1, R36.2, R46.1, R46.2, R47.1, R47.2, U6A.L, U6A.R2, U7.1, U7.2, U7.3. U6.L1/L2 are candidate internal NC pins 4/5; external ties remain unknown. Photo pad S means single pad, not necessarily source. Current completion checklist includes closure criteria and the full per-component audit. Sources: reconstruction.json, remaining_work.json, via_audit.json, v099_net_changes.json, native validation and the preserved earlier measurement records.